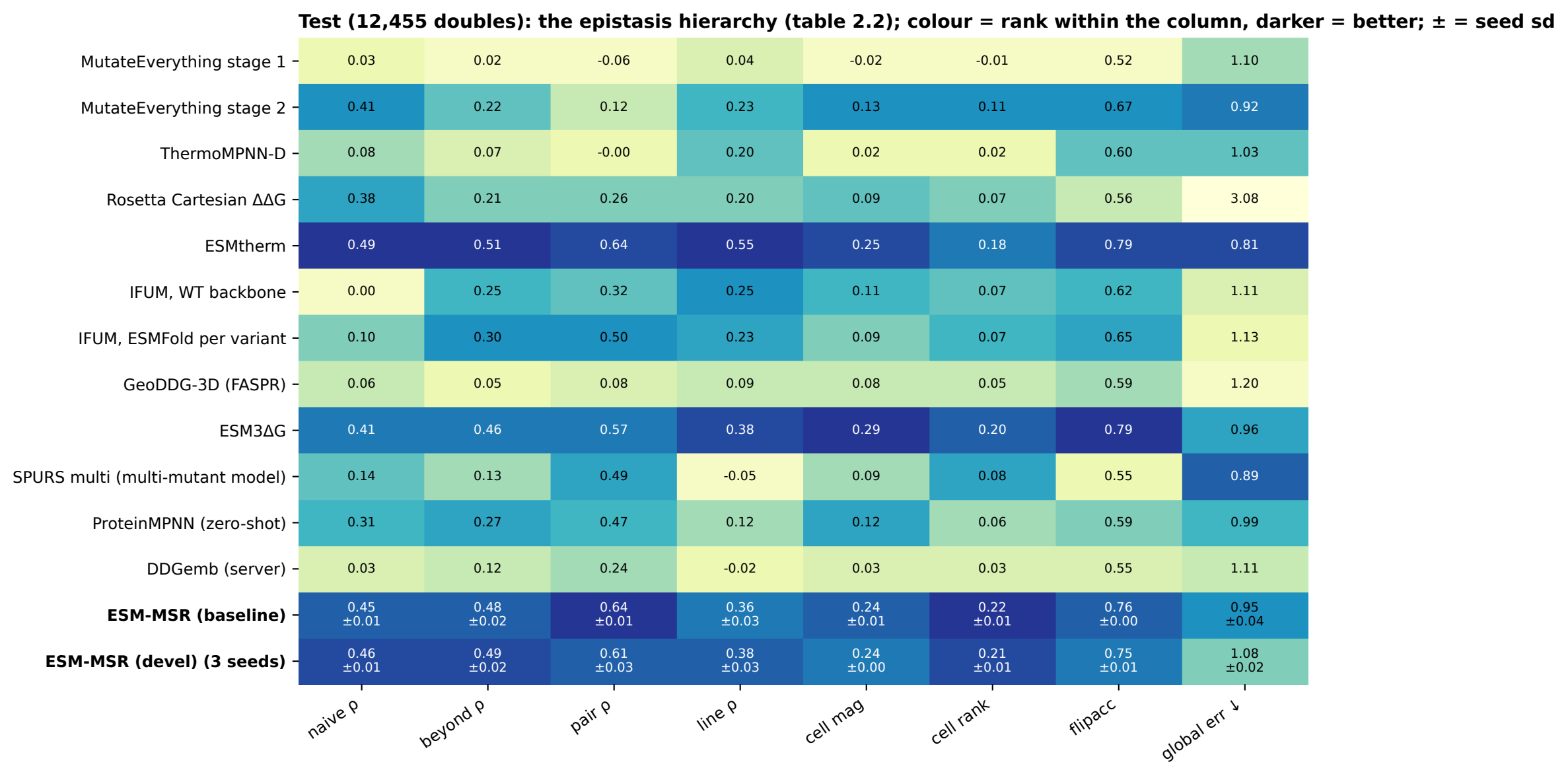
Test (12,455 doubles): the epistasis hierarchy (table 2.2); colour = rank within the column, darker = better; ± = seed sd
MutateEverything stage 1
MutateEverything stage 2
ThermoMPNN-D
Rosetta Cartesian ΔΔG
ESMtherm
IFUM, WT backbone
IFUM, ESMFold per variant
GeoDDG-3D (FASPR)
ESM3ΔG
SPURS multi (multi-mutant model)
ProteinMPNN (zero-shot)
DDGemb (server)
ESM-MSR (baseline)
ESM-MSR (devel) (3 seeds)
0.03 0.02 -0.06 0.04 -0.02 -0.01 0.52 1.10
0.41 0.22 0.12 0.23 0.13 0.11 0.67 0.92
0.08 0.07 -0.00 0.20 0.02 0.02 0.60 1.03
0.38 0.21 0.26 0.20 0.09 0.07 0.56 3.08
0.49 0.51 0.64 0.55 0.25 0.18 0.79 0.81
0.00 0.25 0.32 0.25 0.11 0.07 0.62 1.11
0.10 0.30 0.50 0.23 0.09 0.07 0.65 1.13
0.06 0.05 0.08 0.09 0.08 0.05 0.59 1.20
0.41 0.46 0.57 0.38 0.29 0.20 0.79 0.96
0.14 0.13 0.49 -0.05 0.09 0.08 0.55 0.89
0.31 0.27 0.47 0.12 0.12 0.06 0.59 0.99
0.03 0.12 0.24 -0.02 0.03 0.03 0.55 1.11
0.45 ±0.01 0.48 ±0.02 0.64 ±0.01 0.36 ±0.03 0.24 ±0.01 0.22 ±0.01 0.76 ±0.00 0.95 ±0.04
0.46 ±0.01 0.49 ±0.02 0.61 ±0.03 0.38 ±0.03 0.24 ±0.00 0.21 ±0.01 0.75 ±0.01 1.08 ±0.02
naive ρ
beyond ρ
pair ρ
line ρ
cell mag
cell rank
flipacc
global err ↓